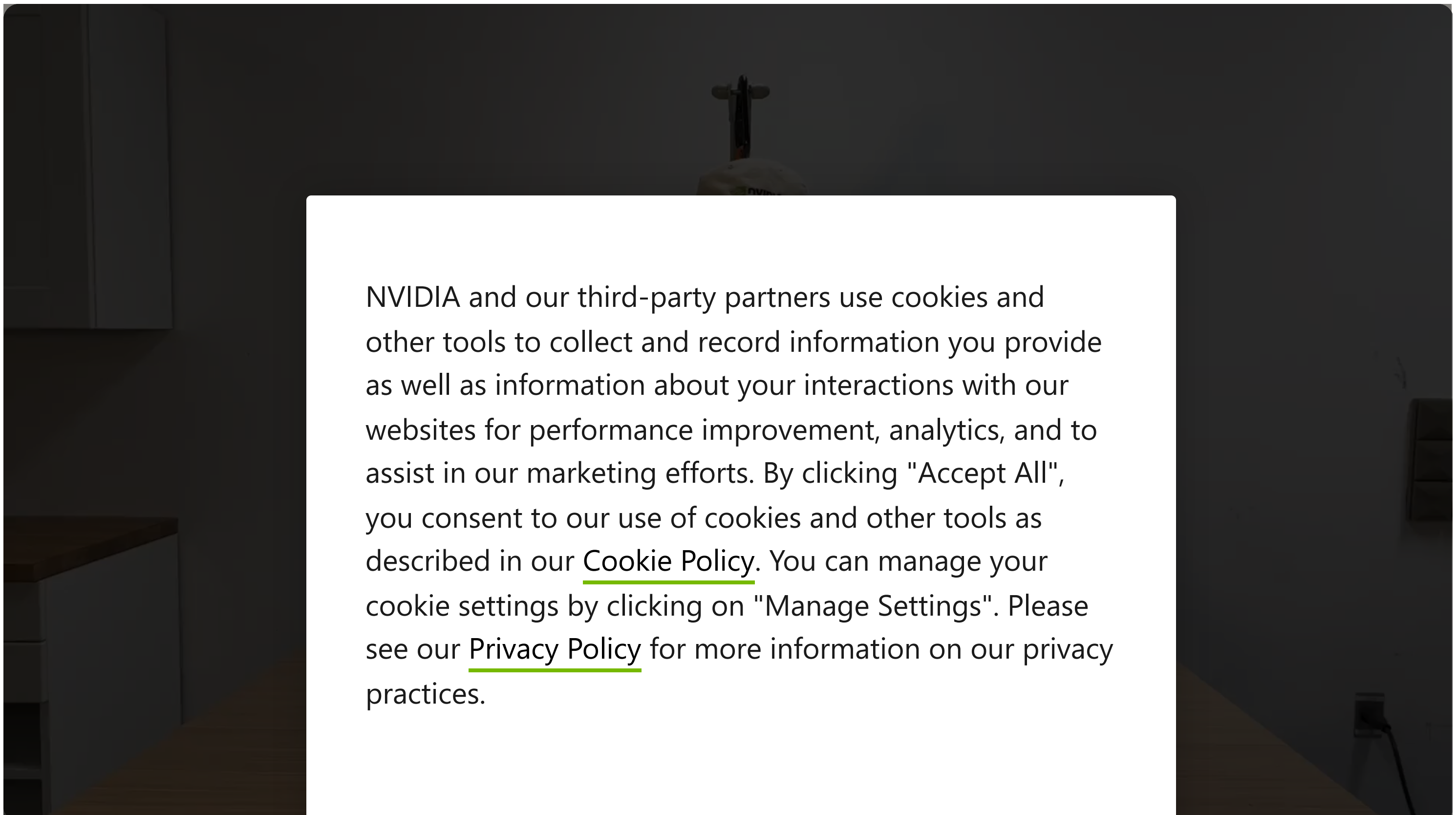
An Improve… …Humanoid



GR00T N1 … o plate"

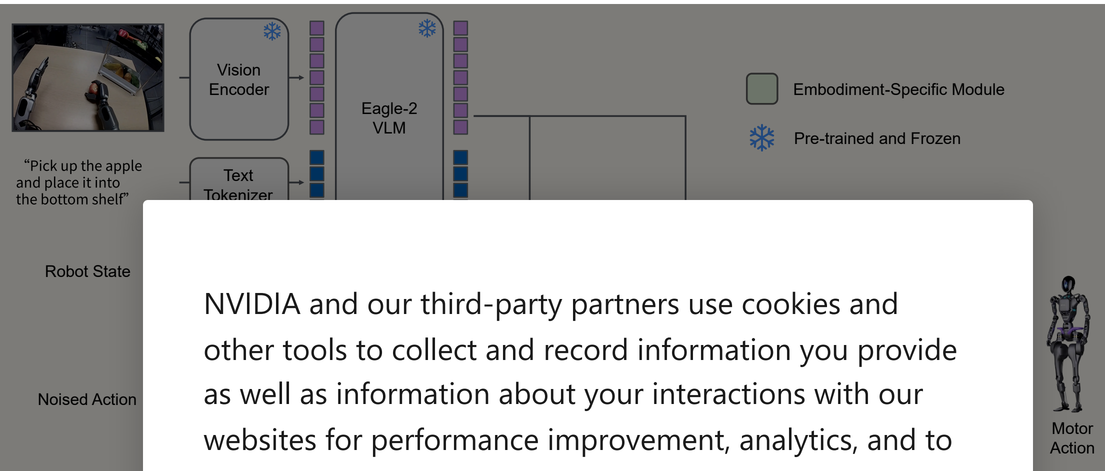
Introduction

We introduce GR00T N1.5, an upgraded version of the GR00T N1 foundation model for humanoid robots. With several architecture, data and modeling improvements, we find that N1.5 outperforms N1 on both simulated manipulation benchmarks and on the real GR-1 robot, detailed below. We expect users of N1.5 should observe better performance compared to N1, in particular improved generalization and better language following ability.

Model and Data Updates

Architecture

As with N1, GR00T N1.5 uses an NVIDIA Eagle VLM to encode text and visual observations. The vision-language embeddings from the VLM are then cross-attended to by the DiT which processes the state and noised actions.



The main differen

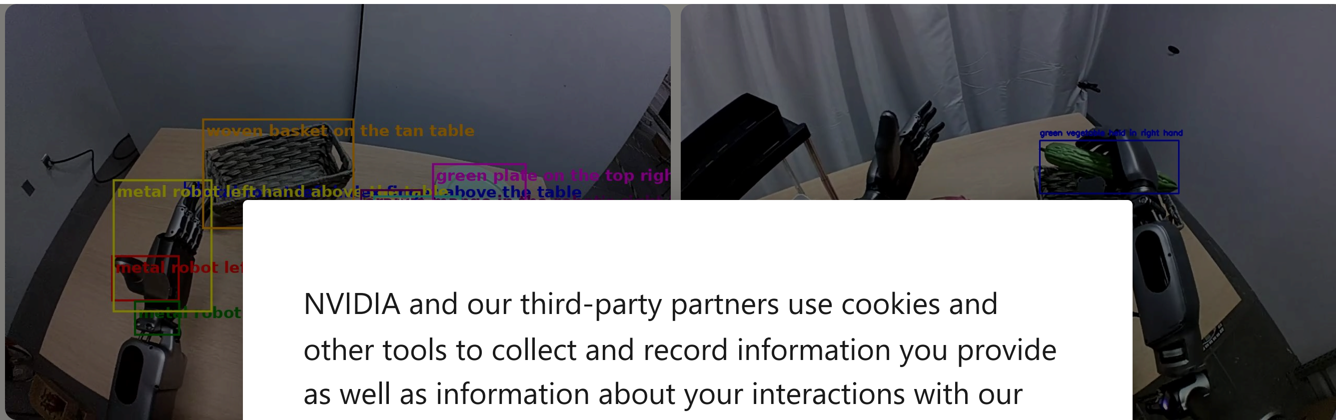
- The VLM model is
- The adapter MLP … dds layer normalization to …

We found that the … eneralization.

Improved VLM Grounding Capabilities

We have updated the VLM of GR00T N1.5, starting from Eagle 2.5 and tuned for better grounding capabilities and physical understanding. On RefCOCOg and our internal GEAR GR-1 grounding dataset with referring expressions, we observe that the N1.5 VLM model performs favorably compared to Qwen2.5-VL-3B - a comparable open-source model.

Model	Size	GR-1 grounding IoU (↑)	RefCOCOg-val IoU (↑)
Qwen2.5VL	3B	35.5	85.2
GR00T N1.5 VLM	2.1B	**40.4**	**89.6**



Left: Ex ... tput.

Joint Policy Le...

In addition to the ... t
Representation Al ... eling future
frames, FLARE ali... dding FLARE
both improves po... n videos.

Training

We trained GR00T ... 16384. As in N1, we used AdamW with cosine learning rate schedule with warmup ratio 0.05. We used FLARE loss coefficient 0.2 for both pretraining and posttraining.

Our pretraining mixture included internal GR-1 data, OpenXE, simulated GR-1 (a.k.a. DexMG), neural trajectories from DreamGen, and AgiBot-Beta:

Experime

Architecture va

In order to tune the model architecture for N1.5, we trained policies from scratch on two sim robot benchmarks requiring language following: Language Table and a set of five simulated GR-1 tasks requring language ("Sim GR-1 Language"). We find that the N1.5 architecture achieves significantly higher success rates on both benchmarks, indicating stronger language-conditioned control ability.

Benchmark	GR00T N1 (scratch)	GR00T N1.5 (scratch)
Language table	52.8%	**93.2%**
Sim GR-1 Language	36.4%	**54.4%**

Data-limited post-training in simulated environments

Following the GR00T N1 evaluation protocol, we evaluate N1.5's performance in data-limited post-training. In the case of Sim GR-1, we can evaluate both fewshot and 0-shot, since the the pretraining mixture includes other Sim GR-1 tasks with the same embodiment. We find that N1.5 is significantly better in the very low data regime (0-shot and 30 demos).

Simulation Benchmark	GR00T N1	GR00T N1.5
RoboCasa, 30 Dem…		
Sim GR-1, 0-shot		
SimGR-1, 30 Demo…		

Real GR-1 langu…

We add a simple l… …s are on a table and the robot is a… …f the target fruit is sampled to be c… …y.

Setting		…1.5
Language followin…		
Overall success rate	[illegible]	[illegible]

We find that N1.5 significantly improves over N1 in terms of its ability to follow language commands on the real GR-1 robot. Although both policies consistently pick and place *some* fruit onto the plate, N1.5 has a much higher language following rate, leading to a higher overall success rate.

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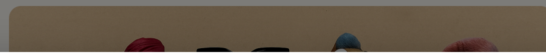
Learning to manipulate novel objects from human ego videos

To evaluate the model's generalization ability, we evaluate pick and place performance using a set of 10 novel objects not seen during pretraining.



As shown in the Fl … s learning directly from hum … cts from human videos an … his also works zero-shot.

← →

Demonstrations using a novel object, captured from a GoPro (left) and the GR-1 robot (right).

Setting	GR00T N1	GR00T N1.5
0-shot	0%	**15.0%**
FLARE post-trained on human videos including novel objects	-	**55.0%**

Novel object generalization performance. We observe that N1.5 both performs better 0-shot, and also benefits from co-training with human videos.

Generalization to novel behaviors using Neural Trajectories

To generalize beyond the teleoperation data and enable humanoid robots to learn new tasks in new environments, we use DreamGen to generate synthetic robot data for training.

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DreamGen 4-step pipeline.

Through the DreamGen pipeline, we show that GR00T N1.5 can achieve non-trivial results on
12 new verbs (see … to the
pretraining data t… tion to new
verbs, only repeat… We find that
GR00T N1.5 achie… 3.1% for GR00T
N1. Although thes… we never
collected teleoper… DreamGen
trajectories; leavi… ure work.

Post-training o…

We post-train GR0… he Unitree G1
robot. As in the Gi… ith one target
object and one di… ty of being
closer to either th… 00T N1.5
achieves much hi… its seen in the
GR-1 pretraining c… unseen objects.

Model	GR00T N1, 1K Demos	GR00T N1.5, 1K Demos	GR00T N1.5, 1K Demos
Task	Place 1 of 2 fruits onto plate; 4 total fruits	Place 1 of 2 fruits onto plate; 4 total fruits	Place 1 of 2 objects onto plate; 5 novel objects
Scene			
Success rate	44.0%	**98.8%**	84.2%

(Left Image) Prompt: Place the bag of chips to the pink plate. (Right Image) Prompt: Place the soap to the blue plate.

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Discussion

Overall, we see that GR00T-N1.5 is a significant improvement over GR00T-N1. It achieves higher success rate, can use more diverse data sources, and has significantly improved language following capabilities. We attribute these improvements to the improved grounding capabilities, usage of the FLARE loss and the diverse data from DreamGen. The model will be open-sourced shortly, and we hope practitioners will observe improved results when finetuning GR00T-N1.5 on their own robots.

Foundation Mo

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